

Spanning R^n

Suppose that we have $v_1, \dots, v_m$ in R^n . How can we tell if they span all of R^n ?

Answer: If we form $A = (v_1 \ v_2 \ \dots \ v_m)$ then $v_1, \dots, v_m$ spans R^n iff when you row reduce the matrix A , every row has a leading 1.

Important fact: If $v_1, \dots, v_m$ spans R^n then $m \geq n$.

A diagram showing a matrix A with columns $v_1, v_2, \dots, v_m$. A vertical double-headed arrow to the left of the matrix is labeled n , indicating the number of rows. A horizontal double-headed arrow below the matrix is labeled m , indicating the number of columns.

$$A = (v_1 \ v_2 \ \dots \ v_m)$$

columns of
the matrix A
($n \times m$).

Question is : If I give you $b \in \mathbb{R}^n$,
is there a solution to $A \underline{x} = b$?
 $\uparrow$
 $m \times 1$ matrix

Ex: $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

all in $\mathbb{R}^3$. Do v_1, v_2 and v_3 span $\mathbb{R}^3$?

Or is there a solution to

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} x = b \quad \text{for all } b \in \mathbb{R}^3.$$

↯ Augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & 1 & b_2 \\ 0 & 0 & 1 & b_3 \end{array} \right)$$

where $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

We get

$$x_3 = b_3$$

$$x_2 = b_2 - b_3$$

$$\begin{aligned} x_1 &= b_1 - (x_2 + x_3) \\ &= b_1 - (b_2 - b_3 + b_3) \\ &= b_1 - b_2. \end{aligned}$$

In general if we row reduce the matrix A and we get a leading 1 in each row then we will know that $Ax = b$ has a solution for all b .

$$\begin{pmatrix} \textcircled{1} & 0 & 0 & 0 \\ 0 & \textcircled{1} & 0 & 0 \\ 0 & 0 & 0 & \textcircled{1} \end{pmatrix}$$

↑

← this is the result
of row reducing.
(A|I).

Ex: $\underline{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

↑ ↑ ↗

3 vectors in $\mathbb{R}^4$.

Do these 3
column vectors
span $\mathbb{R}^4$?

When we row-reduce A we will have at most 3 leading 1's. So for some b , $Ax = b$ will not have a solution.

Ex: $v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ $v_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ $v_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

Claim: v_1, v_2, v_3 does not span $\mathbb{R}^3$.

$v_3 = v_1 + v_2$! v_1 and v_2 don't span $\mathbb{R}^3$

Properties of Multiplication of Matrices and Vectors

linearity.

Theorem

Suppose that A is $n \times m$ and x, y are in R^m . $\rightarrow$ $n \times 1$ vectors

- 1 $A(x + y)$ $= Ax + Ay$ and,
- 2 for every number λ , $A(\lambda x) = \lambda Ax$.

Take a row $r = (r_1 \dots r_m)$ from A and
compute $r(x+y)$ and $rx + ry$.

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} \quad y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

$$r(x+y) = r_1(x_1+y_1) + r_2(x_2+y_2) + \dots + r_m(x_m+y_m)$$

$$= r_1 x_1 + \dots + r_m x_m + r_1 y_1 + \dots + r_m y_m$$

$$= rx + ry$$

But this true in all rows of A so

$$A(x+y) = Ax + Ay.$$

Homogeneous systems

- We say that a system of linear equations is a homogeneous system if all the constants in the equations (the right-hand sides) are 0. A homogeneous system can be written as $Ax = 0$.
- Any homogeneous system of linear equations has at least one solution; that is, $x = 0$.
- Theorem: Any homogeneous system with more unknowns than equations has infinitely many solutions.

$$\text{Ex. } x_1 + 2x_2 + x_3 = 0$$

$$-x_1 - x_2 + x_3 = 0$$

$$2x_1 - x_2 + 3x_3 = 0$$

$$\text{rref} \begin{pmatrix} 1 & 2 & 1 \\ -1 & -1 & 1 \\ 2 & -1 & 3 \end{pmatrix} = \underline{\underline{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}}$$

The only solution is $x = 0$.

Ex:

$$x_1 + 2x_2 + x_3 = 0$$

$$-x_1 - x_2 + x_3 = 0$$

~~row~~ $\left(\begin{array}{ccc} 1 & 2 & 1 \\ -1 & -1 & 1 \end{array} \right) \rightsquigarrow \left(\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array} \right)$

↑
row
reduce.

↑
Free
variable.

Solutions to linear systems using homogeneous equations

Theorem

Suppose that $Ax = b$ is some consistent system of linear equations and x_0 is some solution. Then every solution to $Ax = b$ has the form $x_0 + y$ where y is a solution to the homogeneous equation $Ax = 0$.